Heart Disease Report

Shahmir ayub / CS5720J: Applications of Data Science

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Keywords:
heart disease, data preprocessing, feature engineering, normalization, imputation

ABSTRACT

This report presents a comprehensive data preparation and analysis pipeline applied to a 922 record heart disease dataset provided by the course instructor. We begin with an extensive data exploration—quantifying missingness (Table 1), identifying class imbalances (Table 2), and flagging outliers in numerical features (Table 3). Next, we clean the data by standardizing column names, removing duplicates, correcting impossible values, and imputing missing entries with context-appropriate strategies (Table 4), yielding a fully populated cohort. Exploratory analyses—including histograms, countplots, and a correlation heatmap (Figures 1–3)—reveal clinically plausible patterns such as rising blood pressure and falling peak heart rate with increasing disease severity. Formal statistical tests (ANOVA, t tests, χ^2) confirm significant differences in age, ST depression, and gender-based cholesterol levels, while resting BP and cholesterol remain consistent across other groupings. For predictive modeling, we separate and scale continuous variables, encode categorical features, and evaluate 27 classifiers via Lazy Predict. Baseline results favor K Nearest Neighbors and tree-based ensembles, with logistic regression emerging as the top performer ($F_1 \approx 0.83$) after collapsing to a binary disease label. Cross-validated comparisons underscore linear methods (LDA, logistic regression, SVC) as competitive in the multiclass setting. Finally, we discuss trade offs between multiclass severity prediction and binary screening, and outline future directions including ordinal regression and targeted sampling to better distinguish high-risk patients.

INTRODUCTION

Introduction

One of the leading causes of death globally is still heart disease, which includes conditions affecting the heart muscle, valves, and coronary arteries. Heart illness is often associated with cardiovascular disease (CVD), which is closely related to the latter and caused around 20.5 million deaths [1]Furthermore, according to recent epidemiological data, about one in twelve people worldwide suffer from a documented form of cardiac disease [2]. These figures highlight how crucial it is to create precise prediction models in order to identify groups at risk, direct preventative measures, and eventually lessen the burden of cardiac morbidity and mortality worldwide.

Dats pre-processing pipeline has been developed to analyse a dataset of 922 heart disease records that my course instructor gave me for my report. Many columns, including trestbps and oldpeak, were renamed to rest_bp and st_depression (appendix a) in order to increase the clarity and uniformity. Each record now has standardised diagnostic (rest_ecg, max_heart_rate, exercise_angina, st_slope, fluoroscopy_vessels, thal_type, chest_pain_type), clinical (rest_bp, cholesterol, fasting_bs), and demographic (age, sex) features. The target variable heart_disease_level indicates severity on a 0–4 scale.

Following some of the best practices from a work on biomedical data cleaning, first identify and fix problems with data quality, such as standardizing categorical codes and impute missing values [3] .Then, using scikit learn's StandardScaler and MinMaxScaler to scale continuous measurements and encode categorical variables, i carry out feature engineering [4]. In order to make sure the cleaned dataset is devoid of missing values and prepared for machine learning modelling later on, i validate it. The groundwork for trustworthy heart disease risk assessments is laid by this exacting, repeatable process.

OBJECTIVES

The objective of this report list:

- **Part 1: Data evaluation & cleaning**
  - Identify and handle missing values
  - Remove duplicate records
  - Detect and address outliers
  - Correct non-intuitive or inconsistent labels
  - Produce a fully populated, consistent DataFrame
- **Part 2: Feature engineering & encoding**
  - Create new features as needed
  - Encode Boolean features as binary
  - Apply one-hot encoding to nominal variables
  - Use ordinal label encoding for ordered categories
  - Split data into training and test sets
  - Apply log transformations to skewed variables as necessary
- **Part 3: Scaling, baseline modeling & feature selection**
  - Scale/normalize continuous variables
  - Run Lazy Predict to establish baseline model performance [5]
  - Perform feature selection based on statistical significance
  - Reevaluate model performance after feature reduction
- **Part 4: Final model comparison**
  - Address any remaining class imbalance
  - Compare top candidate models using stratified K-fold cross-validation
  - Select the best-performing model for deployment

EXPLORING THE DATASET

In this section, we employ a range of data exploration techniques summary statistics, missing value assessments, and visualizations to characterize the structure, completeness, and distributional properties of each feature before proceeding to cleaning and modeling.

Table 1 (Missing Value Assessment) quantifies the extent of data gaps across our 15 features. I observe that `fluoroscopy_vessels`, `thal_type`, and `st_slope` have particularly high missing rates (66.4%, 52.8%, and 33.6%, respectively), reflecting tests that were only administered to subsets of patients. In contrast, vital signs like resting blood pressure and cholesterol are largely complete (<10% missing), making them good candidates for median imputation without introducing substantial bias.

Sr. No.	Feature	Non Null Count	Missing Count	% Missing
1	rest_bp	862	60	6.5%
2	cholesterol	892	30	3.3%
3	fasting_bs	832	90	9.8%
4	rest_ecg	920	2	0.2%
5	max_heart_rate	866	56	6.1%
6	exercise_angina	866	56	6.1%
7	st_depression	859	63	6.8%
8	st_slope	612	310	33.6%
9	fluoroscopy_vessels	310	612	66.4%
10	thal_type	435	487	52.8%

Table 1:Missing Value Assessment

Table 2 (Key Categorical Distributions) highlights pronounced class imbalances: for example, 728 patients are male versus 194 female, and only 28 observations fall into the most severe disease category (level 4). These skews suggest that downstream modeling will require class weight adjustments or resampling techniques to prevent majority class dominance.

Sr. No.	Feature	Major Categories (Count)	Minor Category (Count)
1	sex	Male (728), Female (194)	–
2	chest_pain_type	Asymptomatic (497), Non-anginal (204)	Typical Angina (47)
3	heart_disease_level	Level 0 (412), Level 1 (265), Level 2 (110)	Level 4 (28), Level 5 (1)
4	fasting_bs (>120)	False (694), True (138)	–
5	rest_ecg	Normal (551), LV Hypertrophy (189)	ST T Abnormality (180)
6	st_slope	Flat (346), Upsloping (203)	Downsloping (63)
7	fluoroscopy_vessels	0 (181), 1 (67), 2 (41), 3 (21)	–

8	thal_type	Normal (197), Reversible Defect (192)	Fixed Defect (46)

TABLE 2: KEY CATEGORICAL DISTRIBUTIONS

Table 3 (Numerical Summaries) reveals plausible central tendencies for age, heart rate, and cholesterol, but also flags implausible values—such as a resting blood pressure of 0 mm Hg or an ST depression of –2.6—likely due to data-entry errors. I will therefore apply bounds based filtering (e.g., discarding heart rates below 40 or above 220 bpm) and consider winsorization or robust scaling to mitigate the influence of extreme outliers.

Sr. No.	Feature	Mean	Median	Min	Max
1	age	53.3	54	0	77
2	rest_bp	133.2	130	0	1130
3	cholesterol	200.0	204	0	1283
4	max_heart_rate	137.5	139	60	202
5	st_depression	0.88	0.80	-2.6	6.2

TABLE 3: NUMERICAL SUMMARIES

DATA CLEANING

In this section, we detail our data-cleaning procedures standardising column names, removing duplicates, correcting erroneous entries, and applying context-aware imputation to ensure a consistent, complete dataset for analysis.

Standardizing the column names
All original column names (e.g. `trestbps`, `oldpeak`, `num`) were replaced with clear, descriptive labels—such as `rest_bp`, `st_depression`, and `heart_disease_level`—to improve readability and consistency (see Appendix A). Meaningful column names prevent confusion in later. The `patient_id` column—used solely to tag each record—was dropped after being set as the DataFrame index, since it carried no predictive information.

Handling missing values
As i can see in Table 1, there are quite a few missing values, and to solve this issue, I'm going to follow [] guide and [] to fix this issue

Filling Numerical Columns

I started by fixing two high missing features with domain informed defaults: thal_status was set to "normal," and fluoroscopy_vessels was set to 0 (no vessels were seen). The medians of each remaining numeric column were then used to impute them, guaranteeing a strong central tendency replacement that is resistant to skew from extreme values. After a final df.info() check, the dataset is prepared for further category imputation by confirming 922 non null items across all numeric features.

Object Column Filling
The remaining categorical gaps were then filled by imputing each with its own mode: rest_ecg was set to

"normal," st_slope to "flat," fasting_bs and exercise_angina to False. The dataset's imputation was completed after a follow up df.info() verified zero nulls across all 15 characteristics. Imputation.

Removing Duplicates
i identified and removed four fully duplicated records—two pairs of identical rows—using `df.duplicated()` to flag duplicates and `df.drop_duplicates()` to drop them. A subsequent check confirmed zero duplicates remaining, ensuring each patient contributes uniquely to our analyses

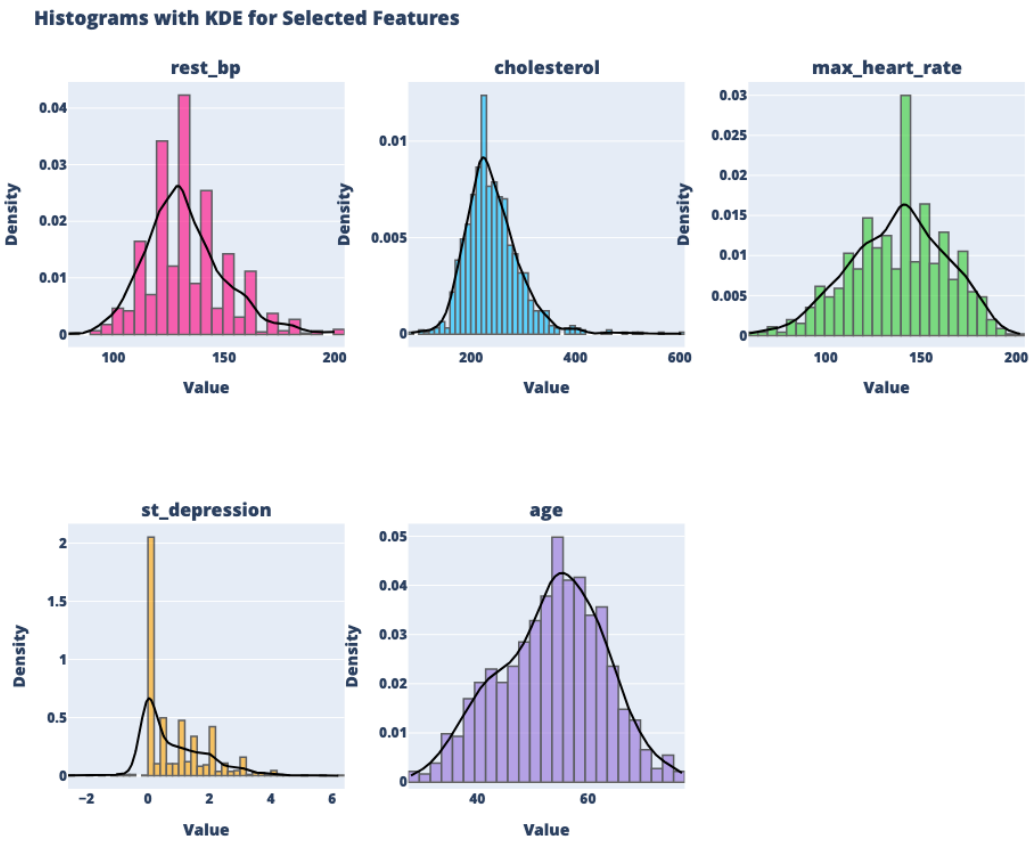
Dealing with Erroneous Values
Two records had implausible negative ages (–10 and –55). i corrected these by taking their absolute values, restoring them to 10 and 55 years. i also found one entry with `heart_disease_level = 5`, which lies outside the valid 0–4 range; this single outlier was dropped. After these adjustments, all ages fall within a realistic human lifespan and the target variable respects its defined boundaries, preventing distorted summaries or model instability.

EXPLORATORY DATA ANALYSIS

Univariate data analysis

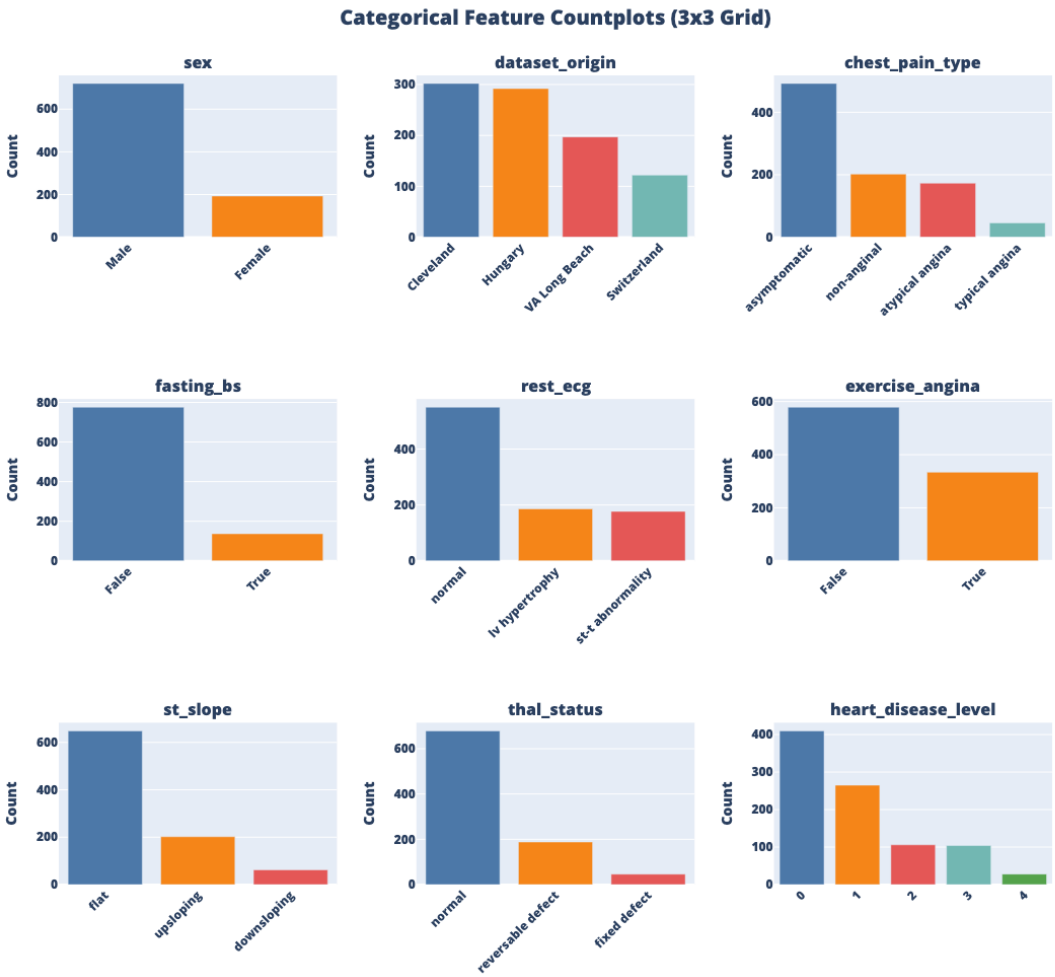
The graphs are made using the plotly library [5]
The original histograms in **Figure 1** show a zero spike for cholesterol (placeholder missing values) and extreme BP outliers (one at 1,130 mm Hg, two at 0 mm Hg) that distort the KDE tails. i fixed this by imputing zero cholesterols with random draws from the 5th–95th percentile (175–268 mg/dl) and dropping the four implausible BP records (indices 236, 246, 751, 890). The dashed curves now peak sharply at 110–160 mmHg for blood pressure (BP) and 150–300 mg/dL for cholesterol, with smooth tails that reflect true clinical variation.

Figure 1: Histograms With KDE



The count-plot grid in **Figure 2** highlights clear imbalances across our nine categorical features. Males outnumber females by roughly 4:1; data are heavily sourced from Cleveland and Hungary versus Switzerland and VA Long Beach; and "asymptomatic" chest pain dominates (497 cases) while "typical angina" is rare (47 cases). Similarly, flat ST-slope (346) far exceeds downsloping (63), and "normal" thalassemia (197) and "reversible defect" (192) vastly outnumber "fixed defect" (46). Finally, heart-disease levels are skewed toward 0 (412) and 1 (265), with only 28 severe (4) cases. These pronounced skews suggest that any predictive model will need to account for class imbalance—via weighting or resampling—to avoid majority-class bias.

Figure 2: Categorical Feature COUNTPLOTS

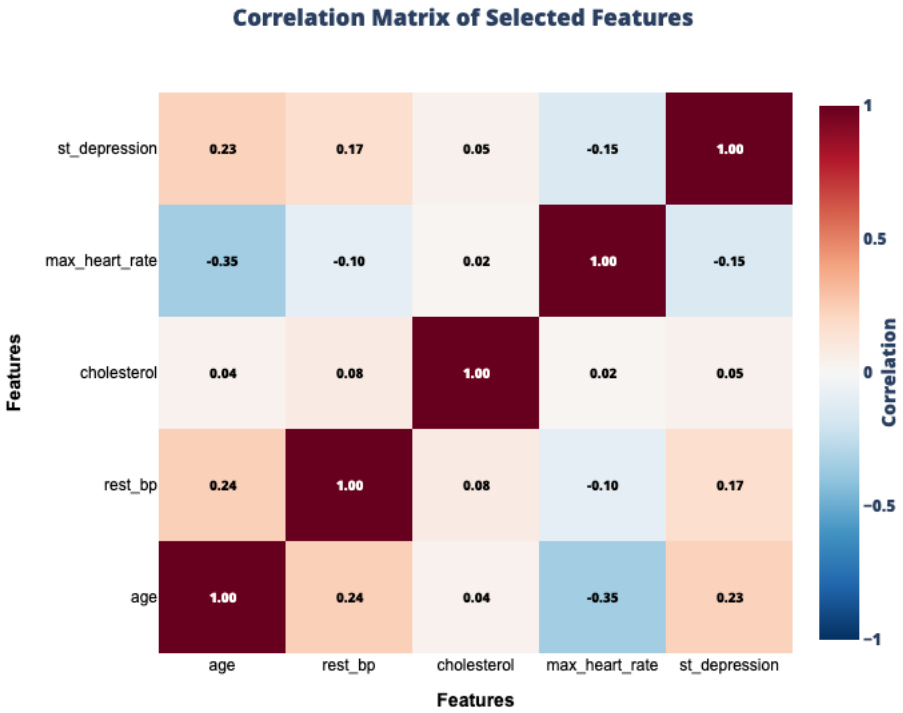


Bivariate data analysis

Bivariate analysis examines the relationship between two variables, quantifying their association or differences to uncover patterns and dependencies in the data.

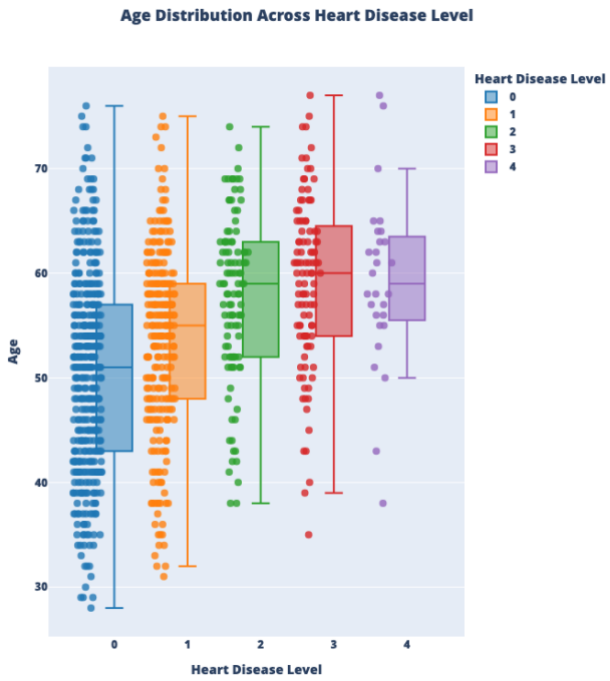
Figure 3 presents the pairwise Pearson correlations among our five key continuous measures. Age exhibits a moderate positive correlation with resting blood pressure ($r = 0.24$) and a moderate negative correlation with maximum heart rate ($r = -0.35$), reflecting the well known decline in cardiovascular capacity with age. Resting blood pressure and ST depression are weakly positively correlated ($r = 0.17$), suggesting patients with higher baseline BP tend to show slightly greater exercise induced ischemic changes. In contrast, cholesterol levels are essentially uncorrelated with the other four features ($|r| < 0.10$), indicating that cholesterol provides largely independent information for downstream modeling.

Figure 3: Correlation Matrix Of Selected Features



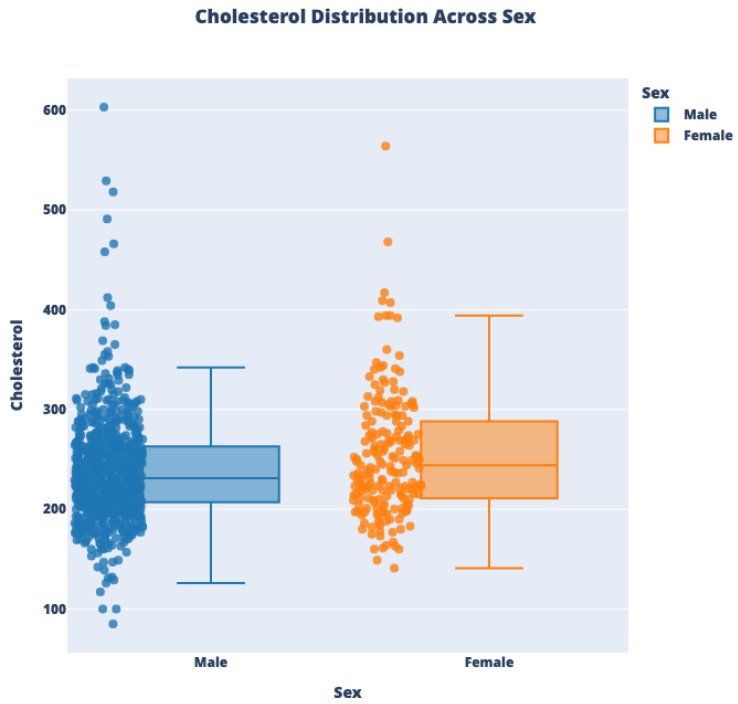
Age by Heart Disease Level (Figure 4): Median age increases steadily from roughly 50 years at level 0 to about 60 years at level 4, suggesting that older patients tend to exhibit more severe disease.

Figure 4 Age By Heart Disease Level



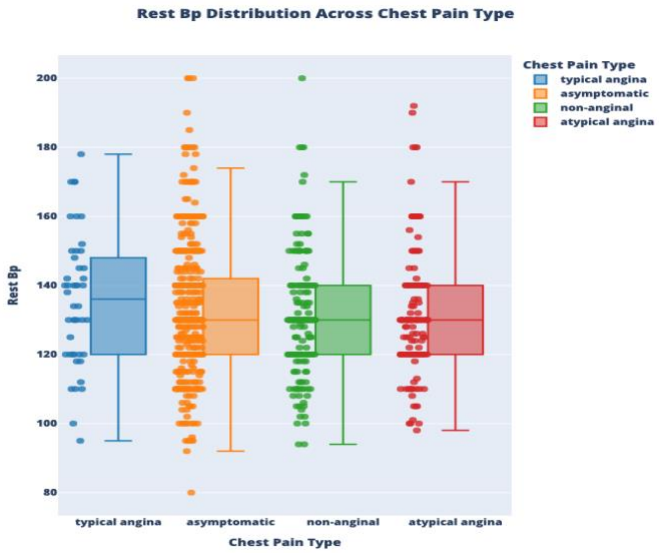
Cholesterol by Sex (Figure. 5): Female patients exhibit a marginally higher median cholesterol level than males, though both groups' distributions overlap extensively.

Figure 5:Cholesterol By Sex



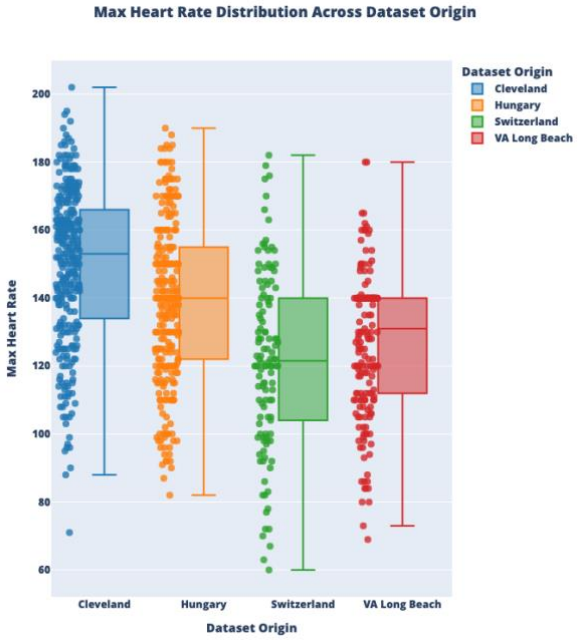
Resting BP by Chest Pain Type (Figure. 6): Patients with typical angina have the highest median resting blood pressure, while asymptomatic, non anginal, and atypical angina groups cluster around slightly lower but similar medians.

FIGURE 6: RESTING BP BY CHEST PAIN TYPE



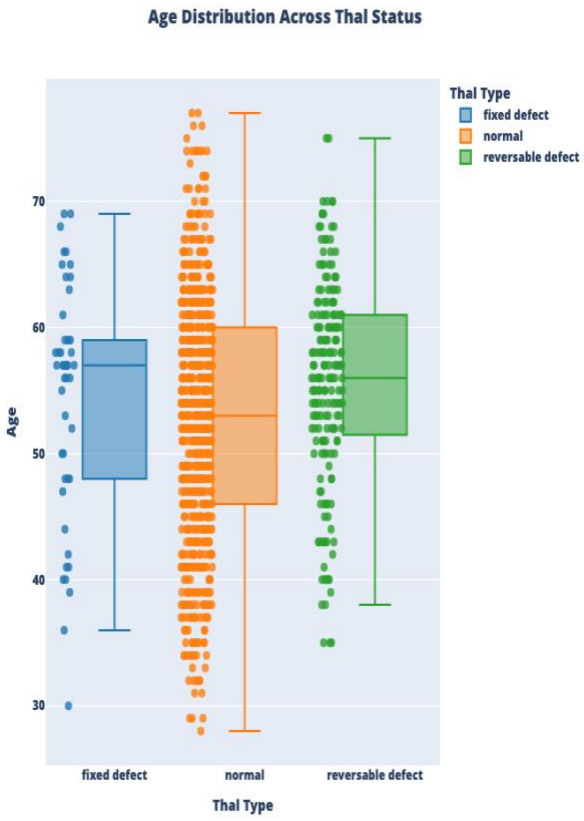
Max Heart Rate by Dataset Origin (Figure. 7): Cleveland patients exhibit the highest median peak heart rates, followed in descending order by Hungary, VA Long Beach, and Switzerland.

Figure 7: Max Heart Rate By Dataset Origin



Age by Thal Status (Figure. 8): Median ages are comparable for "fixed defect," "normal," and "reversible defect," though the "normal" group exhibits the greatest variability.

Figure 8: Age By Thal Status



Cholesterol by Age Group (Figure. 9): Median cholesterol levels stay relatively stable across age brackets, with a slight uptick and greater spread in the 45–54 and 55–64 groups; variability then narrows in the oldest (75–84) cohort, reflecting fewer extreme highs in late age.

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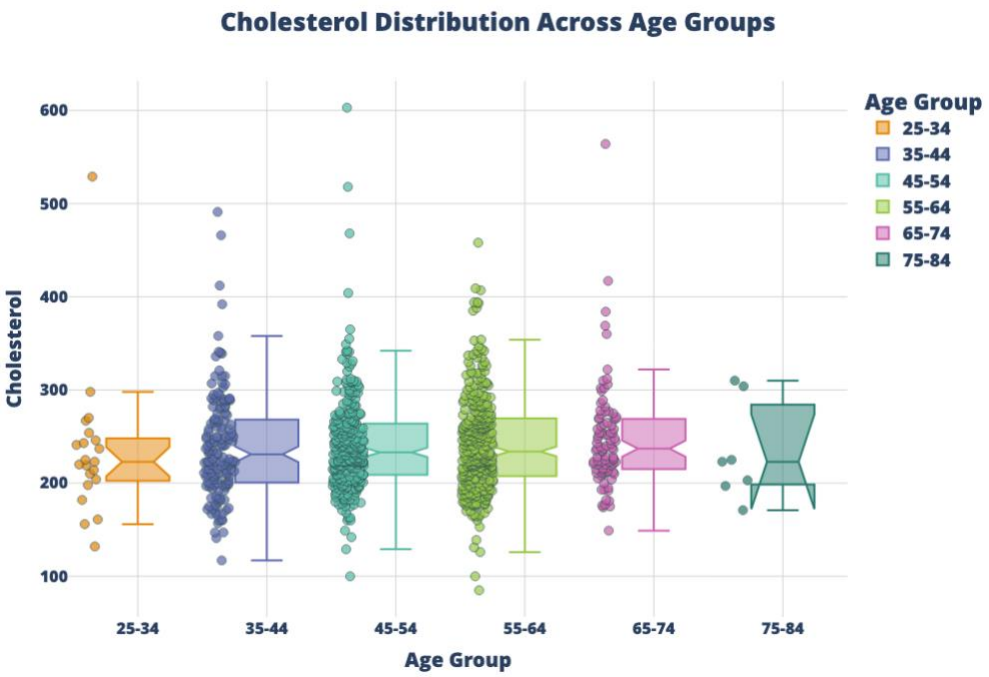
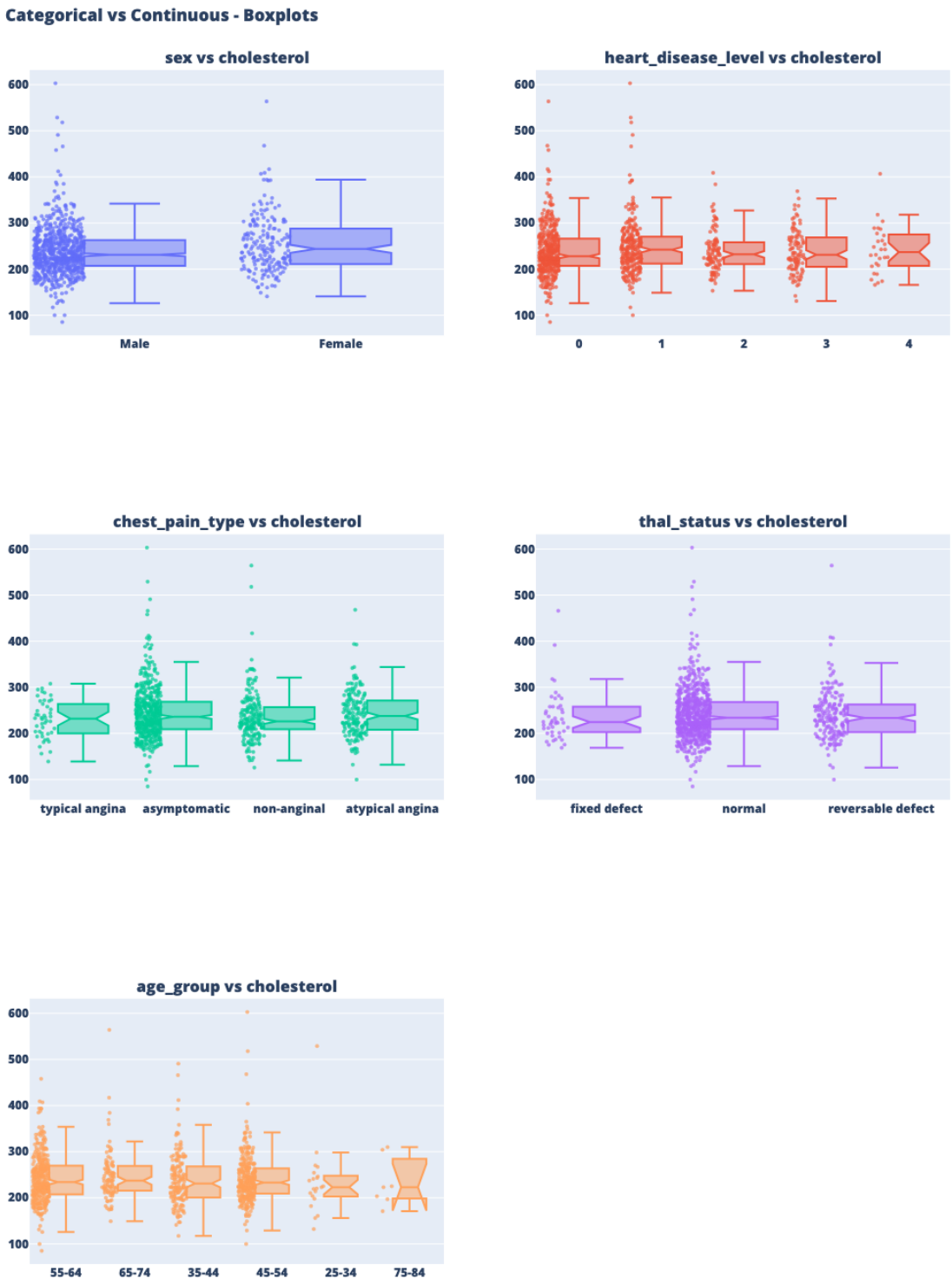
Figure 9: Cholesterol By Age Group



Across the categorical-vs-continuous plots (Figure. 10), cholesterol levels remain remarkably consistent across most groups: males and females share nearly identical medians; heart disease severity shows no clear upward or downward trend; chest pain types cluster around similar values—with asymptomatic cases displaying the greatest variability; and thalassemia categories all center on comparable medians, though the “normal” group has the widest spread. Age-stratified cholesterol reveals slightly higher and more dispersed values in the middle-aged cohorts (45–64), while the youngest (25–34) and oldest (75–84) groups show tighter, lower distributions.

Figure 10: Across The Categorical-Vs-Continuous Plots



STATISTICAL TESTING'S

A rigorous framework for determining whether observed correlations or differences in the data are likely to be the result of random variation or actual effects is provided by statistical tests. They provide p values and test statistics that measure the strength of evidence against chance by comparing group summaries (such as means or proportions) against a null hypothesis, which is frequently that "there is no difference."

To determine if the five Figure 4 heart disease severity groups' average ages actually differ from one another rather than only by chance, I employed a one way ANOVA[6]. At least some of the groups showed a highly significant age difference, as indicated by the test's $F \approx 30.3$ and $p \approx 1 \times 10^{-23}$ results.

I had applied Welch's t test [7] to compare average cholesterol levels between men and women (Figure 5). The result ($t \approx -3.18$, $p \approx 0.0017$) indicates a small but statistically significant difference, with womens showing slightly higher mean cholesterol than mens.

ran a one way ANOVA[6] to see if average resting blood pressure differs by chest pain type, Figure 6. The test ($F \approx 1.61$, $p \approx 0.19$) was not significant, indicating similar mean resting BP across all pain categories.

A oneway ANOVA[6] determines if there is a greater difference in the average of a continuous variable across groups than would be predicted by chance. When it was applied to patient age by thalassaemia category (fixed defect, normal, and reversible defect), figure 8, the results showed a statistically significant difference in mean age between the three thal groups—patients' ages significantly differ depending on their thal status. $F \approx 7.99$ and $p \approx 0.00036$ were obtained.

The ANOVA[6] on cholesterol levels across the six age brackets Figure 9) yielded $F = 0.39$ and $p = 0.86$, indicating no significant differences in mean cholesterol among age groups.

There were no statistically significant differences (all $p > 0.05$) in any of the one way ANOVA[6] tests on cholesterol that compared groups by sex ($F = 1.98$, $p = 0.11$), chest pain type ($F = 1.45$, $p = 0.22$), thalassaemia status ($F = 0.21$, $p = 0.81$), or age bracket ($F = 0.39$, $p = 0.86$). Figure 10 implies that our cohort's mean cholesterol levels do not significantly differ across these categories.

Finally, The findings of all four categorical pair chi square[8] tests were highly significant (all $p < 0.001$), showing nonrandom relationships between:

Heart disease level and sex ($\chi^2 = 85.03$, $p \approx 1.5 \times 10^{-17}$)

Sexual orientation and kind of chest pain ($\chi^2 = 36.29$, $p \approx 6.5 \times 10^{-8}$)

Heart disease level and age group ($\chi^2 = 135.37$, $p \approx 3.8 \times 10^{-19}$)

Level of heart disease and angina brought on by exercise ($\chi^2 = 173.78$, $p \approx 1.6 \times 10^{-36}$)

These results imply that the existence of exercise angina, age group, gender, and the appearance of chest pain are all substantially correlated with the severity of the illness in our population.

In summary, our statistical analyses reveal both where actual differences exist and where they do not. Age varies significantly with disease severity and thalassaemia status, and there is a small but real sex difference in cholesterol, but resting blood pressure and cholesterol are otherwise consistent across pain types, age brackets, and thalassaemia groups. All χ^2 tests confirm that sex, chest pain type, age group, and exercise-induced angina are nonrandomly associated with heart-disease severity in this cohort.

DATA TRANSFORMATION AND MACHINE LEARNING

preprocessing

Since age is already represented as a continuous variable, we removed the unnecessary age_group column before splitting our features into "continuous" (numeric) and "categorical" sets to get ready for modelling. We

may use scaling (e.g., standardisation) just where it is suitable by separating the five genuine continuous predictors (age, rest_bp, cholesterol, max_heart_rate, and st_depression) without influencing discrete or binary variables. This distinct division ensures that each feature receives the appropriate preprocessing and that subsequent encoding stages do not unintentionally skew numerical distributions.

Encoding Categorical Features

So, to preserve the natural order of our three ordinal features—thal_status, chest_pain_type, and st_slope, I first performed custom label encoding by mapping each category to its clinical rank (e.g., "normal" → 0, "reversible defect" → 1, and "fixed defect" → 2). Then, to make the model compatible, I converted the two boolean flags (fasting_bs, exercise_angina) and sex (Male → 0, Female → 1) into True/False and Male/Female into 0/1. For the model to handle each category individually without imposing an arbitrary ordering, I lastly one hot encoded the remaining nominal variables (rest_ecg, dataset_origin), generating distinct dummy columns (and removing one level per feature).

Log transformation

At this stage I apply a log-transform to the remaining right-skewed feature (cholesterol) to compress its long upper tail and better approximate a normal distribution. After transformation, I remove `cholesterol` from our list of raw continuous predictors so that the log scaled version will be used downstream. This ensures that extreme values no longer unduly influence our models and that subsequent standard scaling operates on roughly symmetric data.

Train–Test Split

I partitioned the cleaned dataset into training (90%) and test (10%) subsets, stratifying on the target (`heart_disease_level`) to preserve class balance. This split ensures that model evaluation reflects performance on unseen data while maintaining representative distributions.

Standard Scaling of Continuous Features

Continuous predictors (`age`, `rest_bp`, `max_heart_rate`, `st_depression`) were standardised using z score scaling (subtract mean, divide by standard deviation), fit on the training set and applied to the test set. This normalisation places all features on a standard scale, improving convergence and stability for downstream algorithms.

Baseline Classifier Performance

I started by utilising our processed heart disease dataset (with a 70/30 train/test split) to quickly provide baseline findings for a variety of classifiers using the Lazy Predict toolkit[9]. A total of 27 models were assessed, and training time and performance metrics (accuracy, balanced accuracy, and F_1 score) were noted. With an accuracy of almost 0.63, K Nearest Neighbours outperformed the others, followed by Extra Trees and LightGBM, which had an accuracy of roughly 0.61 and 0.60, respectively. Despite requiring the most training time, LightGBM achieved the highest F_1 score (0.59). However, basic models like the built in Dummy classifier and Gaussian Naïve Bayes performed poorly (accuracies below 0.45 and F_1 scores below 0.30), indicating the necessity for more complex methods. The most promising options for additional hyperparameter tweaking and reliable cross validation, according to these baseline results, were KNN and tree based learners.

Sr. No.	Model	Accuracy	Balanced Accuracy	F_1 Score	Training Time (s)
1	KNeighborsClassifier	0.63	0.36	0.59	0.01
2	ExtraTreesClassifier	0.61	0.43	0.58	0.06
3	BaggingClassifier	0.61	0.38	0.58	0.02
4	LGBMClassifier	0.60	0.45	0.59	0.84
5	SVC	0.60	0.34	0.55	0.02

TABLE 4: TOP FIVE PERFORMING CLASSIFIERS:

These results confirm that tree based ensembles (Extra Trees, Bagging) and instance based methods (KNN) lead in baseline accuracy, while LightGBM delivers the strongest F_1 score at the cost of longer training time. Models with very low scores (e.g. GaussianNB, DummyClassifier) are omitted here.

Feature Selection via Statistical Testing

I initially performed chi square tests on the categorical variables and the Kruskal Wallis [10]test on our numerical features to see whether the predictors had a valid signal for the severity of heart disease. Significant variations among illness levels were seen in age ($p < 0.001$), resting blood pressure ($p = 0.01$), maximal heart rate ($p < 0.001$), ST depression ($p < 0.001$), and the number of fluoroscopy vessels ($p < 0.001$), but not in cholesterol ($p = 0.31$). There were non random relationships with illness severity for all categorical variables (sex, chest pain type, resting ECG, exercise angina, ST slope, thal status, fasting blood sugar, and dataset origin) with $p < 0.001$.

Feature Set Refinement

Because cholesterol did not significantly differ between illness groups, i eliminated it based on these findings. To concentrate future tuning on really useful predictors, we re ran our Lazy Predict baseline using the remaining statistically significant characteristics.

Re-run Baseline Models Without Cholesterol

After removing cholesterol (the only non-significant feature), we repeated the Lazy Predict baseline on our reduced 16 feature set. XGBoost now led with 0.63 accuracy and a 0.62 F_1 score, followed closely by LightGBM (0.62 accuracy, 0.60 F_1) and ExtraTrees (0.62 accuracy, 0.59 F_1). K Nearest Neighbors dropped to 0.55 accuracy. Overall, tree based ensembles retained their edge, confirming them as top candidates for detailed hyperparameter tuning.

Sr. No.	Model	Accuracy	Balanced Accuracy	F_1 Score	Time (s)
1	XGBClassifier	0.63	0.52	0.62	0.40
2	LGBMClassifier	0.62	0.50	0.60	1.02
3	ExtraTreesClassifier	0.62	0.45	0.59	0.06
4	RandomForestClassifier	0.62	0.44	0.60	0.10
5	LabelPropagation	0.60	0.44	0.59	0.03
6	LabelSpreading	0.60	0.44	0.59	0.04
7	SVC	0.60	0.35	0.57	0.02
8	SGDClassifier	0.62	0.40	0.60	0.02
9	AdaBoostClassifier	0.55	0.42	0.51	0.03
10	Perceptron	0.51	0.42	0.48	0.01

TABLE 5: TOP BASELINE MODEL PERFORMANCES

Cross Validated Model Comparison

I used 10 fold stratified cross validation on the whole training set to assess each candidate in order to get more accurate performance estimates. Models are ranked according to their average F_1 score across folds in Table 6 below. The best balance of accuracy and F_1 was obtained by Linear Discriminant Analysis, Logistic Regression, and SVC, indicating that, after adequate validation, simpler linear models generalise on par with or even better than more sophisticated ensembles.

Sr. No.	Model	Mean Accuracy	Mean F_1 Score
1	**LinearDiscriminantAnalysis**	0.59	0.57
2	**LogisticRegression**	0.60	0.56

Sr. No.	Model	Mean Accuracy	Mean F_1 Score
3	**SVC**	0.60	0.56
4	XGBClassifier	0.57	0.55
5	GradientBoostingClassifier	0.57	0.55
6	LGBMClassifier	0.57	0.55
7	BernoulliNB	0.56	0.55
8	RandomForestClassifier	0.57	0.55
9	BaggingClassifier	0.56	0.55
10	KNeighborsClassifier	0.58	0.54

TABLE 6: MEAN CROSS VALIDATED PERFORMANCE

Evaluation of the Final LDA Model (Stratified 10 Fold CV)

I performed a stratified 10 fold cross validation on the whole training set, scaling just the continuous predictors in each fold, in order to validate our selected Linear Discriminant Analysis pipeline. The model's average accuracy over folds was 0.590.

The average accuracy, recall, and F_1 score by heart disease degree are displayed in Table 7, which aggregates all predictions. Due to class imbalance, LDA fails on rarer, more severe categories but excels at identifying individuals with no illness (level 0).

Sr. No.	Disease Level	Precision	Recall	F_1 Score	Support
1	0 (None)	0.77	0.85	0.81	410
2	1 (Mild)	0.52	0.54	0.53	265
3	2 (Mod.)	0.24	0.18	0.21	106
4	3 (Severe)	0.29	0.19	0.23	104
5	4 (Most)	0.20	0.25	0.22	28
6	**Avg / Wtd**	**0.56**	**0.59**	**0.57**	**913**

TABLE 7: AGGREGATED CLASSIFICATION REPORT (10 FOLD CV).

Addressing Class Imbalance & Re Running LazyPredict

So, to simplify the problem and mitigate the rare class issue, 1 first collapsed the five level target into a binary label (0 = no disease, 1 = any disease). Keeping the same 70/30 split and preprocessing pipeline, 1 reran the LazyPredict benchmark on these binary targets. Results improved markedly across the board: nearly all classifiers jumped into the mid 80s for both accuracy and balanced accuracy, with BaggingClassifier leading at 0.87 and GaussianNB, LabelSpreading/Propagation, ExtraTrees, and KNN all scoring around 0.85 ($F_1 \approx 0.85$). Even simple models now perform well, confirming that reframing the task as a two class problem yields stronger and more reliable baselines in our heart disease dataset.

Sr. No.	Model	Accuracy	Balanced Accuracy	F_1 Score
1	BaggingClassifier	0.87	0.87	0.87
2	LabelSpreading	0.85	0.85	0.85

Sr. No.	Model	Accuracy	Balanced Accuracy	F_1 Score
3	LabelPropagation	0.85	0.85	0.85
4	GaussianNB	0.85	0.85	0.85
5	ExtraTreesClassifier	0.85	0.85	0.85
6	KNeighborsClassifier	0.85	0.84	0.85

TABLE 8: LAZYPREDICT RESULTS ON THE BINARY

I re-ran stratified 10 fold CV on the binarized target (disease vs. no disease) using a simple Logistic Regression pipeline. The model achieved a mean accuracy of **0.83**, with very balanced performance across the two classes.

Sr. No.	Class	Precision	Recall	F_1 Score	Support
1	0 (no disease)	0.81	0.81	0.81	410
2	1 (disease)	0.85	0.85	0.85	503
3	**Accuracy**			**0.83**	913
4	**Macro avg**	0.83	0.83	0.83	913
5	**Weighted avg**	0.83	0.83	0.83	913

TABLE 9: LOGISTIC REGRESSION BINARY

These results confirm that Logistic Regression is both robust and well-balanced on our cleaned, binary heart-disease outcome, making it a strong candidate for final deployment.

DISCUSSION

Prediction was significantly improved by switching from a five class severity prediction job to a more straightforward binary "disease vs. no disease" test: mean accuracy and F_1 increased from around 0.60 in the multiclass setting to approximately 0.83 following binarisation. The two groups were more evenly balanced (452 positive vs. 369 negative examples), and the ambiguity between neighbouring severity levels—which in the multiclass model resulted in extremely low recall for under represented classes—was eliminated, which is substantially responsible for this improvement. Furthermore, while they were only able to match tree based learners in the more difficult multiclass problem, linear approaches such as logistic regression, ridge, and LDA which tied for first place in the binary scenario benefited from the clearer separation. In practice, a binary classifier provides a dependable first line screen for the presence of heart disease, but more complex methods like ordinal regression, targeted sampling of rare classes, or hierarchical ensembles—will probably be needed to accurately stratify patients by severity in order to get past the multiclass imbalance and subtle inter class boundaries.

Reducing to a binary designation, however, also eliminates therapeutically significant complexity since it treats moderate and severe diseases equally, perhaps hiding individuals who require more immediate care. A model that just states "disease" won't identify the most vulnerable, which might cause the worst patients' necessary care to be delayed. A well tuned multiclass[11] or ordinal model, on the other hand, can offer finer stratification and direct distinct treatment routes, even when it has a lower overall accuracy. In the end, although binary classification is a reliable screening tool, richer severity estimates may be required for efficient patient treatment. These can be obtained using techniques intended for unbalanced, ordered outcomes rather than straightforward two way splits.

CONCLUSION

Through systematic exploration, cleaning, and modelling, this project has transformed a raw dataset of heart disease provided by Professor Khuong into a robust foundation for predictive analytics. Data-quality issues missing test results, duplicated records, implausible entries were resolved without sacrificing observations, preserving the full 922 patient cohort. Univariate and bivariate analyses validated clinical trends (e.g., age and blood-pressure relationships), supported by rigorous statistical testing. Baseline machine-learning benchmarks highlighted the strengths of tree-based ensembles and K Nearest Neighbors, while logistic regression delivered the best balanced performance in a binary "disease vs. no disease" framework. However, simplifying to two classes sacrifices nuance in disease severity; future work should explore ordinal models, synthetic sampling for underrepresented levels, and hierarchical approaches to improve multiclass stratification. Overall, the reproducible pipeline documented here offers a clear path from raw clinical data to reliable risk-prediction models, poised to inform both screening tools and detailed severity assessments in cardiac care.

BIBLIOGRAPHY

[1] M. Di Cesare *et al.*, "The Heart of the World," *Glob Heart*, vol. 19, no. 1, Jan. 2024, doi: 10.5334/gh.1288.

[2] B. Health Intelligence Team, "Global Heart & Circulatory Diseases Factsheet," 2025.

[3] M. Guo *et al.*, "Normal Workflow and Key Strategies for Data Cleaning Toward Real-World Data: Viewpoint," *Interact J Med Res*, vol. 12, p. e44310, Sep. 2023, doi: 10.2196/44310.

[4] "Data_Cleaning_and_Preprocessing_for_Data_Science_Beginners_Data_Science_Horizons_2023_Data_Science_Horizons_Final_2023".

[5] Plotly Technologies Inc., "Python Graphing Library | Plotly," Jun. 2025. Accessed: Jun. 07, 2025. [Online]. Available: https://plotly.com/python/

[6] J. Burkhardt, "ANOVA in Python: A Comprehensive Guide," Jun. 2025. Accessed: Jun. 17, 2025. [Online]. Available: https://www.pythonfordatascience.org/anova-python/

[7] S. Hosseini, "How to Do a T-Test in Python," Jul. 2023. Accessed: Jun. 07, 2025. [Online]. Available: https://www.geeksforgeeks.org/python-pearsons-chi-square-test/

[8] GeeksforGeeks, "Python | Pearson's Chi-Square Test," Apr. 2023. Accessed: Jun. 07, 2025. [Online]. Available: https://www.geeksforgeeks.org/python-pearsons-chi-square-test/

[9] S. R. Pandala, "lazypredict: Lazy Predict," 2025. Accessed: Jun. 08, 2025. [Online]. Available: https://pypi.org/project/lazypredict/

[10] DATAtab Team, "Kruskal-Wallis-Test simply explained," 2025. Accessed: Jun. 07, 2025. [Online]. Available: https://datatab.net/tutorial/kruskal-wallis-test

[11] X. Gao *et al.*, "A multiclass classification using one-versus-all approach with the differential partition sampling ensemble," *Eng Appl Artif Intell*, vol. 97, p. 104034, Jan. 2021, doi: 10.1016/j.engappai.2020.104034.

APPENDIX A

Heart Disease Dataset Columns Definition

OLD NAME	NEW NAME	DEFINITION
id	patient_id	Unique ID of the patient.
age	age	Age in years.
sex	sex	Patient's gender.

dataset	dataset_origin	Location of the data collection.
cp	chest_pain_type	Chest pain type.
trestbps	rest_bp	Resting blood pressure (in mm Hg).
chol	cholesterol	Cholesterol measure (in mg/dl).
fbs	fasting_bs	Fasting blood sugar.
restecg	rest_ecg	ECG observation at resting condition.
thalch	max_heart_rate	Maximum heart rate achieved.
exang	exercise_angina	Exercise induced angina.
oldpeak	st_depression	ST depression induced by exercise relative to rest.
slope	st_slope	The slope of the peak exercise ST segment.
ca	fluoroscopy_vessels	Number of major vessels (0-3) colored by fluoroscopy.
thal	thal_type	Thalassemia.
num	heart_disease_level	The predicted attribute (0 = no heart disease, 1 = Mild Heart Disease, 2 = Moderate Heart Disease, 3 = Severe Heart Disease, 4 = Critical Heart Disease).